

```
// Experiment 5: Controlling DC Motor Using Transistor Driver Circuit and  
Display potentiometer value and motor speed on 16x2 I2C LCD
```

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
const int Pot = A0;
```

```
const int motorA1 = 5;
```

```
const int motorA2 = 6;
```

```
const int enableA = 9;
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    pinMode(Pot, INPUT);
```

```
    pinMode(motorA1, OUTPUT);
```

```
    pinMode(motorA2, OUTPUT);
```

```
    pinMode(enableA, OUTPUT);
```

```
    lcd.init();
```

```
    lcd.backlight();
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("Pot: ");
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Speed: ");  
}  
  
void loop() {  
    int potVal = analogRead(Pot);  
    int speed = map(potVal, 0, 1023, 0, 255);  
  
    digitalWrite(motorA1, HIGH);  
    digitalWrite(motorA2, LOW);  
    analogWrite(enableA, speed);  
  
    Serial.print("Potentiometer Value: ");  
    Serial.print(potVal);  
    Serial.print(" | Motor Speed (PWM): ");  
    Serial.println(speed);  
  
    lcd.setCursor(5, 0);        // after "Pot: "  
    lcd.print(potVal);  
    lcd.print("  ");  
    lcd.setCursor(7, 1);  
    lcd.print(speed);  
    lcd.print("  ");  
  
    delay(200);  
}
```